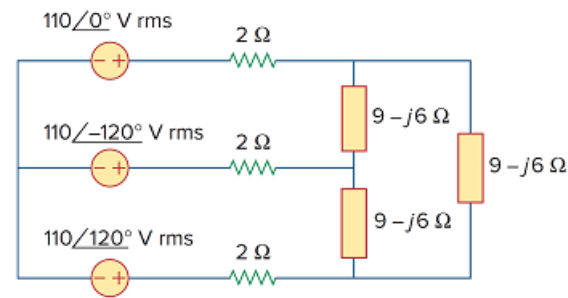


Problem

In the balanced three-phase Y- Δ system in Fig. 12.46, find the line current I_L and the average power delivered to the load.

Figure 12.46



Step-by-step solution

Step 1 of 4

Refer to the circuit diagram in Figure 12.46 in the textbook.

Convert $\text{Y-}\Delta$ circuit into Y-Y circuit by dividing Δ load by 3.

$$Z_Y = \frac{Z_{\Delta}}{3}$$

Substitute $9 - j6$ for Z_{Δ} in the equation.

$$\begin{aligned} Z_Y &= \frac{9 - j6}{3} \\ &= (3 - j2) \, \Omega \end{aligned}$$

Step 2 of 4

Replace Δ load of Figure 12.46 with Y load as shown in Figure 1.

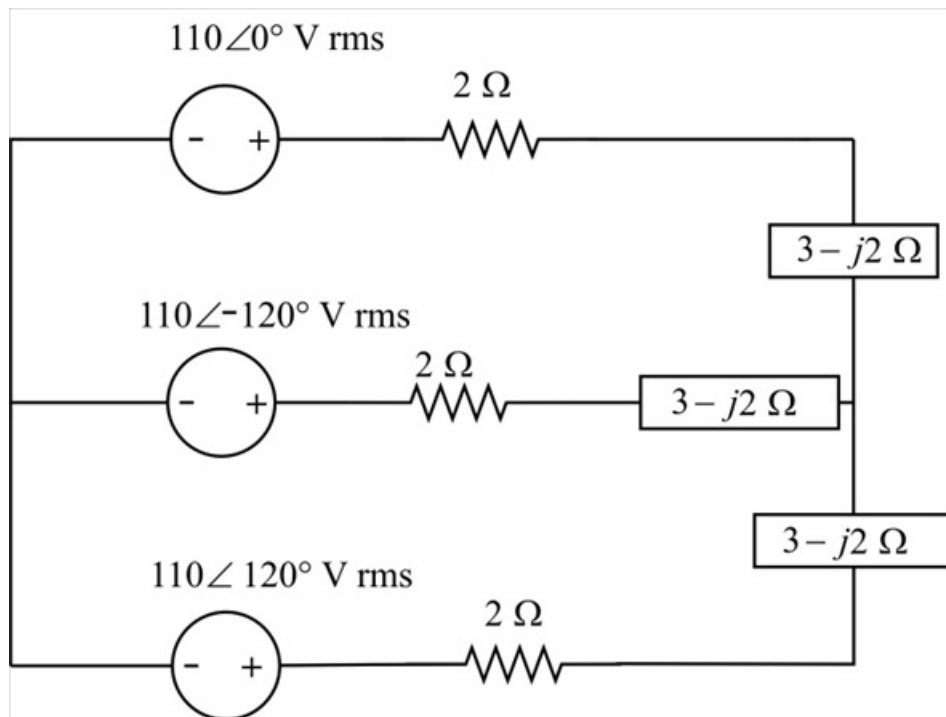


Figure 1

Figure is the Y - Y system and is balanced. An alternative way of analyzing a balanced Y - Y system is into a single phase equivalent circuit as shown in Figure 2.

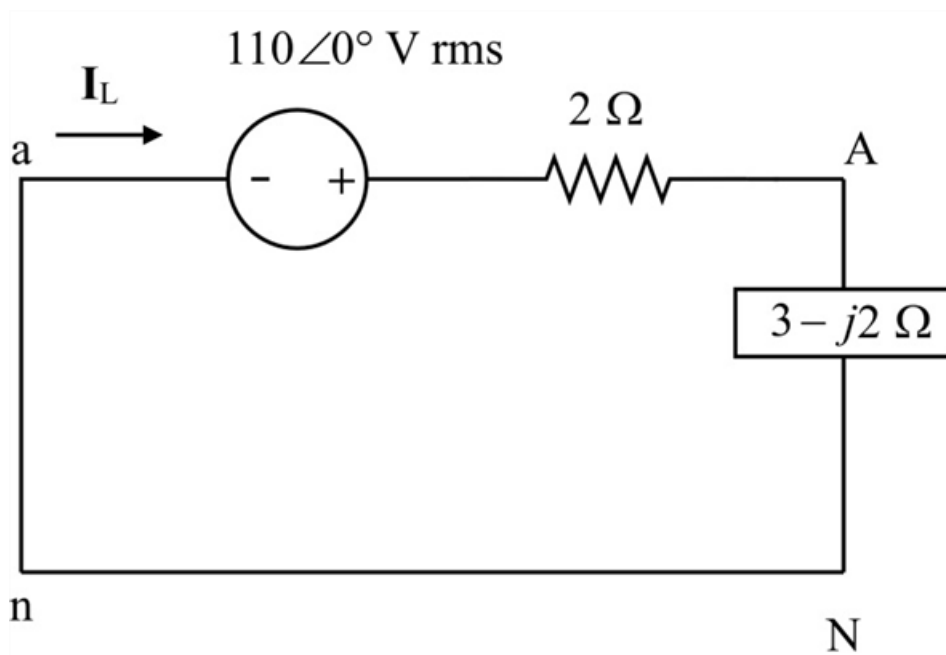


Figure 2

Step 3 of 4

From figure 2, For Y - Y system the line current I_L is equals to I_p .

$$\begin{aligned} I_L &= \frac{110\angle 0^\circ \text{ V}}{2 + 3 - 2j} \\ &= \frac{110\angle 0^\circ \text{ V}}{5 - 2j} \\ &= 18.96 + 7.58j \\ &= 20.42\angle 21.80^\circ \text{ A} \end{aligned}$$

Thus, the line current I_L is $\boxed{20.42 \text{ A}}$.

Step 4 of 4

Find the average power delivered to the load.

$$P = 3(I_L')^2 R'$$

Substitute 20.42 for I_L' , 3 for R' in the equation.

$$\begin{aligned} P &= 3(20.42)^2 (3) \\ &= 3752.78 \text{ Watts} \end{aligned}$$

Thus, power P delivered to the load is **3752.78 Watts**.